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DGMSpecification CSISSurveillanceDGM {
  description "Dynamic goals for the CSIS campus surveillance system at UFBA"
  operationalMode NORMAL
  DynamicGoal InterpretWeaponDetection {
     title "Interpreting weapon detection across varying campus contexts"
     useCaseRef "UC_DetectWeapon"
     ObservableAction {
        signature "WeaponDetected(weaponType, cameraID, location, t, person)"
        trigger "YOLOv11x weapon detection model identifies a firearm or bladed
weapon in frame"
        scopeStart "Detection model returns positive weapon classification with
confidence >= 0.75"
        scopeEnd "Operator confirms or dismisses the alert on the Dashboard"
     }
     Evidence WeaponClassified {
        type DERIVED
        source "YOLOv11x weapon detection model"
        predicate "WeaponClassified(weaponType, confidence) where confidence >=
0.75"
     }
     Evidence PersonInUniform {
        type IDENTITY
        source "YOLOv11x person detection model with appearance classification"
        predicate "PersonWearingUniform(person) == true"
     }
     Evidence LocationIsSensitive {
        type SPATIAL
        source "camera location registry"
        predicate "LocationType(cameraID) IN {classroom, laboratory, library,
administration}"
     }
     Evidence LocationIsEntrance {
        type SPATIAL
        source "camera location registry"
        predicate "LocationType(cameraID) IN {mainGate, parkingEntrance,
perimeterGate}"
     }
     Evidence IsAfterHours {
        type TEMPORAL
        source "system clock and campus schedule"
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      predicate "CurrentTime NOT IN businessHours"
    }
    Evidence HighConfidence {
      type SENSOR
      source "YOLOv11x weapon detection model"
      predicate "DetectionConfidence >= 0.90"
    }
    GoalHypothesis CriticalArmedThreat {
      intention "An unauthorized individual is carrying a weapon in a sensitive campus
location, posing an immediate threat to the academic community"
      condition WeaponClassified && NOT PersonInUniform && LocationIsSensitive
&& HighConfidence
      ResponseBinding {
        action "Generate CRITICAL alert, notify Supervisor immediately, contact
Federal Police, dispatch all available SecurityAgents to location"
        ucmStep "UC_DetectWeapon.Step4"
      }
    }
    GoalHypothesis PerimeterWeaponConcern {
      intention "A weapon is detected at a campus entrance or perimeter, possibly an
individual attempting to bring a weapon onto campus"
      condition WeaponClassified && NOT PersonInUniform && LocationIsEntrance
      ResponseBinding {
        action "Generate HIGH alert, notify Operator and Supervisor, dispatch nearest
SecurityAgent to verify, restrict entry at affected gate"
        ucmStep "UC_DetectWeapon.Step4"
      }
    }
    GoalHypothesis AuthorizedArmedPersonnel {
      intention "A uniformed law enforcement officer or authorized security personnel
is carrying a service weapon as part of their duties"
      condition WeaponClassified && PersonInUniform && LocationIsEntrance
      ResponseBinding {
        action "Log detection event for record keeping, no alert generated, mark as
routine observation"
        ucmStep "UC_DetectWeapon.Step3a"

      }
    }
    GoalHypothesis AmbiguousWeaponSighting {
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        intention "A weapon-like object is detected but context is insufficient to
determine threat level with confidence"
        condition WeaponClassified && NOT HighConfidence
        ResponseBinding {
          action "Generate MEDIUM alert, request Operator to review footage manually,
flag for Supervisor attention"
          ucmStep "UC_DetectWeapon.Step3a"
        }
      }
      precedence Precedence {
        justification "Immediate threat to life in a sensitive location takes highest priority.
Perimeter detection is next because it represents a potential escalation. Ambiguous
detection requires human review but is not an emergency. Authorized personnel
sighting is lowest priority."
        CriticalArmedThreat > PerimeterWeaponConcern > AmbiguousWeaponSighting
> AuthorizedArmedPersonnel
      }
      escalation Escalation {
        strategy NOTIFY_OPERATOR
        additionalEvidence PersonInUniform, LocationIsSensitive, HighConfidence
        defaultResponse "Treat as potential threat: generate HIGH alert and dispatch
SecurityAgent to verify in person"
        timeout "30s"
      }
    }
    DynamicGoal InterpretSuspiciousBehaviorNearVehicle {
      title "Interpreting person behavior near vehicles in parking areas"
      useCaseRef "UC_DetectSuspiciousBehavior"
      ObservableAction {
        signature "PersonNearVehicle(personID, vehiclePlate, posture, duration,
cameraID, t)"
        trigger "YOLOv11x-Pose model detects a person in sustained proximity to a
vehicle in a parking area"
        scopeStart "Person tracked within 2 meters of a vehicle for more than 30
seconds"
        scopeEnd "Person moves away from vehicle or Operator resolves the alert"
      }
      Evidence PersonCrouching {
        type DERIVED
        source "YOLOv11x-Pose keypoint analysis"
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    predicate "BodyPosture(personID) == CROUCHING"
  }
  Evidence PersonStanding {
    type DERIVED
    source "YOLOv11x-Pose keypoint analysis"
    predicate "BodyPosture(personID) == STANDING"
  }
  Evidence DurationExceedsShortThreshold {
    type TEMPORAL
    source "person tracking across frames"
    predicate "ProximityDuration(personID, vehiclePlate) >= 60s"
  }
  Evidence DurationExceedsLongThreshold {
    type TEMPORAL
    source "person tracking across frames"
    predicate "ProximityDuration(personID, vehiclePlate) >= 120s"
  }
  Evidence IsAfterHours {
    type TEMPORAL
    source "system clock and campus schedule"
    predicate "CurrentTime NOT IN businessHours"
  }
  Evidence VehicleFlagged {
    type CATALOG
    source "vehicle restriction database"
    predicate "VehicleRestrictionStatus(vehiclePlate) == FLAGGED"
  }
  Evidence VehicleAuthorized {
    type CATALOG
    source "vehicle registration database"
    predicate "VehicleAuthorized(vehiclePlate) == true"
  }
  Evidence ConfidenceAboveThreshold {
    type SENSOR
    source "YOLOv11x-Pose model"
    predicate "DetectionConfidence >= 0.80"
  }
  GoalHypothesis VehicleTheftAttempt {
    intention "An individual is attempting to break into, damage, or steal a vehicle in
the parking area"
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        condition PersonCrouching && DurationExceedsShortThreshold &&
IsAfterHours && ConfidenceAboveThreshold
        ResponseBinding {
          action "Generate CRITICAL alert with severity high, start 20-second video
recording, notify SecurityAgent and Supervisor immediately"
          ucmStep "UC_DetectSuspiciousBehavior.Step5"
        }
      }
      GoalHypothesis SuspiciousLoitering {
        intention "An individual is loitering near vehicles after business hours in a
manner that warrants investigation"
        condition PersonStanding && DurationExceedsLongThreshold && IsAfterHours
&& ConfidenceAboveThreshold
        ResponseBinding {
          action "Generate HIGH alert with severity medium, start video recording,
notify SecurityAgent for on-site assessment"
          ucmStep "UC_DetectSuspiciousBehavior.Step5"
        }
      }
      GoalHypothesis FlaggedVehicleInteraction {
        intention "A person is interacting with a vehicle that has been flagged in the
restriction database, regardless of time of day"
        condition (PersonCrouching || PersonStanding) &&
DurationExceedsShortThreshold && VehicleFlagged
        ResponseBinding {
          action "Generate HIGH alert, cross-reference plate data with incident history,
notify Supervisor with combined plate and behavior data"
          ucmStep "UC_DetectSuspiciousBehavior.Step5"
        }
      }
      GoalHypothesis RoutineActivity {
        intention "A person is loading or unloading items from their own authorized
vehicle during normal business hours"
        condition PersonStanding && VehicleAuthorized && NOT IsAfterHours
        ResponseBinding {
          action "Log observation for audit trail, no alert generated"
          ucmStep "UC_DetectSuspiciousBehavior.Step5"
        }
      }
      precedence Precedence {
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        justification "Active break-in attempt with crouching posture after hours is the most urgent. Standing loitering after hours is next. Interaction with a flagged vehicle is concerning regardless of posture or time. Routine activity with an authorized vehicle during business hours is benign."
        VehicleTheftAttempt > SuspiciousLoitering > FlaggedVehicleInteraction > RoutineActivity
      }
      escalation Escalation {
        strategy NOTIFY_OPERATOR
        additionalEvidence IsAfterHours, VehicleFlagged, PersonCrouching
        defaultResponse "Treat as suspicious: generate MEDIUM alert, dispatch SecurityAgent for visual verification"
        timeout "60s"
      }
  }
  DynamicGoal InterpretSmokeDetection {
      title "Interpreting smoke or fire detection across different campus locations"
      useCaseRef "UC_DetectFireSmoke"
      ObservableAction {
        signature "SmokeOrFireDetected(detectionType, cameraID, location, t)"
        trigger "YOLOv11x fire and smoke detection model identifies smoke or fire in frame"
        scopeStart "Detection model returns positive fire or smoke classification"
        scopeEnd "Operator confirms or dismisses the alert, or emergency services respond"
      }
      Evidence FireClassified {
        type DERIVED
        source "YOLOv11x fire and smoke detection model"
        predicate "DetectionClass == FIRE"
      }
      Evidence SmokeClassified {
        type DERIVED
        source "YOLOv11x fire and smoke detection model"
        predicate "DetectionClass == SMOKE"
      }
      Evidence PersistsAcrossFrames {
        type TEMPORAL
        source "consecutive frame analysis"
        predicate "DetectionPersistence >= 5 consecutive frames"
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    }
    Evidence LocationIsKitchenArea {
      type SPATIAL
      source "camera location registry"
      predicate "LocationType(cameraID) IN {cafeteria, kitchen, foodCourt}"
    }
    Evidence LocationIsLaboratory {
      type SPATIAL
      source "camera location registry"
      predicate "LocationType(cameraID) IN {chemistryLab, physicsLab,
researchLab}"
    }
    Evidence LocationIsOpenArea {
      type SPATIAL
      source "camera location registry"
      predicate "LocationType(cameraID) IN {courtyard, field, parking, walkway}"
    }
    Evidence HighConfidence {
      type SENSOR
      source "YOLOv11x fire and smoke detection model"
      predicate "DetectionConfidence >= 0.80"
    }
    GoalHypothesis ActiveFireEmergency {
      intention "An active fire is burning on campus and requires immediate
emergency response to protect lives and property"
      condition FireClassified && PersistsAcrossFrames && HighConfidence
      ResponseBinding {
        action "Generate CRITICAL alert, switch to FireEmergency mode, notify all
Operators and Supervisor, contact FireDepartment, dispatch all SecurityAgents to
evacuate affected area"
        ucmStep "UC_HandleFireEmergency.Step1"
      }
    }
    GoalHypothesis LaboratorySmokeHazard {
      intention "Smoke detected in a laboratory may indicate a chemical reaction,
equipment failure, or fire in a hazardous environment requiring specialized response"
      condition SmokeClassified && LocationIsLaboratory && PersistsAcrossFrames
      ResponseBinding {
        action "Generate HIGH alert, notify Supervisor and laboratory safety officer,
dispatch SecurityAgent to assess, prepare for possible evacuation of building"
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            ucmStep "UC_DetectFireSmoke.Step4"
        }
    }
    GoalHypothesis KitchenSmokeEvent {
        intention "Smoke detected near a kitchen or food preparation area is likely from cooking activity and not a fire emergency"
        condition SmokeClassified && LocationIsKitchenArea && NOT FireClassified
        ResponseBinding {
            action "Generate LOW alert for Operator review, log event, no automatic escalation unless fire is subsequently detected"
            ucmStep "UC_DetectFireSmoke.Step3a"
        }
    }
    GoalHypothesis OutdoorSmokeSource {
        intention "Smoke detected in an open area may be from debris burning, vehicle exhaust, or an external source, requiring investigation but not immediate emergency response"
        condition SmokeClassified && LocationIsOpenArea && NOT FireClassified
        ResponseBinding {
            action "Generate MEDIUM alert, request Operator to review footage, dispatch SecurityAgent if smoke persists beyond 5 minutes"
            ucmStep "UC_DetectFireSmoke.Step4"
        }
    }
    precedence Precedence {
        justification "Active fire confirmed across multiple frames is the most critical scenario requiring immediate evacuation and emergency services. Laboratory smoke is next due to chemical hazard potential. Outdoor smoke requires investigation. Kitchen smoke is likely benign and lowest priority."
        ActiveFireEmergency > LaboratorySmokeHazard > OutdoorSmokeSource > KitchenSmokeEvent
    }
    escalation Escalation {
        strategy NOTIFY_OPERATOR
        additionalEvidence PersistsAcrossFrames, FireClassified, HighConfidence
        defaultResponse "Treat as potential fire: generate HIGH alert, notify Supervisor, dispatch SecurityAgent for immediate on-site assessment"
        timeout "15s"
    }
}
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DynamicGoal InterpretUnregisteredPlate {
    title "Interpreting an unregistered vehicle license plate at campus entrance"
    useCaseRef "UC_DetectLicensePlate"
    ObservableAction {
        signature "UnregisteredPlateDetected(plateText, cameraID, t, confidence)"
        trigger "PaddleOCR recognizes a plate not found in the university vehicle
database"
        scopeStart "Plate recognition returns text with confidence >= 0.85 and database
query returns no match"
        scopeEnd "Operator resolves the alert or vehicle exits campus"
    }
    Evidence PlateNotInDatabase {
        type DERIVED
        source "vehicle registration database query"
        predicate "VehicleAuthorized(plateText) == false"
    }
    Evidence PlateSeenFrequently {
        type TEMPORAL
        source "event history database"
        predicate "AppearanceCount(plateText, last30Days) >= 5"
    }
    Evidence PlateSeenFirstTime {
        type TEMPORAL
        source "event history database"
        predicate "AppearanceCount(plateText, last30Days) == 1"
    }
    Evidence PlateInStolenDatabase {
        type CATALOG
        source "vehicle restriction database"
        predicate "VehicleRestrictionStatus(plateText) == STOLEN"
    }
    Evidence IsVisitorHours {
        type TEMPORAL
        source "system clock and campus schedule"
        predicate "CurrentTime IN visitorHours"
    }
    Evidence HighOCRConfidence {
        type SENSOR
        source "PaddleOCR"
        predicate "OCRConfidence >= 0.95"
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    }
    GoalHypothesis StolenVehicleOnCampus {
      intention "A vehicle reported as stolen has entered the campus, requiring immediate law enforcement response"
      condition PlateNotInDatabase && PlateInStolenDatabase && HighOCRConfidence
      ResponseBinding {
        action "Generate CRITICAL alert, notify Supervisor and Operator immediately, contact FederalPolice with plate data and entry location, dispatch SecurityAgent to track vehicle"
        ucmStep "UC_VerifyVehicleAuthorization.Step3"
      }
    }
    GoalHypothesis FrequentUnauthorizedVehicle {
      intention "An unregistered vehicle has been appearing on campus repeatedly, suggesting it should either be registered or investigated"
      condition PlateNotInDatabase && PlateSeenFrequently && NOT PlateInStolenDatabase
      ResponseBinding {
        action "Generate MEDIUM alert, compile appearance history for Supervisor review, recommend adding plate to watch list or registration"
        ucmStep "UC_VerifyVehicleAuthorization.Step3"
      }
    }
    GoalHypothesis LikelyVisitor {
      intention "An unregistered vehicle entering during visitor hours for the first time is likely a legitimate visitor"
      condition PlateNotInDatabase && PlateSeenFirstTime && IsVisitorHours && NOT PlateInStolenDatabase
      ResponseBinding {
        action "Log entry for record keeping, no alert generated"
        ucmStep "UC_VerifyVehicleAuthorization.Step3a"
      }
    }
    GoalHypothesis UnknownVehicleAfterHours {
      intention "An unregistered unknown vehicle entering campus outside visitor hours warrants attention"
      condition PlateNotInDatabase && PlateSeenFirstTime && NOT IsVisitorHours && NOT PlateInStolenDatabase
      ResponseBinding {
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            action "Generate LOW alert for Operator review, log entry with timestamp and camera location"
            ucmStep "UC_VerifyVehicleAuthorization.Step3"
          }
        }
        precedence Precedence {
          justification "A stolen vehicle on campus is an immediate safety concern requiring police. A frequently appearing unregistered vehicle suggests a pattern that needs investigation. An unknown vehicle after hours is worth noting. A first-time visitor during business hours is almost certainly benign."
          StolenVehicleOnCampus > FrequentUnauthorizedVehicle > UnknownVehicleAfterHours > LikelyVisitor
        }
        escalation Escalation {
          strategy REQUEST_MORE_EVIDENCE
          additionalEvidence PlateInStolenDatabase, PlateSeenFrequently, HighOCRConfidence
          defaultResponse "Log detection and generate LOW alert for Operator review"
          timeout "120s"
        }
    }
}
```